Roll No. | | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 18

B.Tech. (CSE) (Sem.–5)

COMPUTATIONAL BIOLOGY

Subject Code : BTCS 521-18

M.Code : 78327

Time : 3 Hrs. Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.
3. SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.

SECTION-A

Write briefly :

1) Difference between aerobic and anaerobic respiration.

2) What are sense and anti-sense codons?

3) What do you mean by TrEMBL and PROSITE?

4) Explain Viterbi algorithms for sequence analysis.

5) Energetics of krebs cycle.

6) Classify proteins.

7) Basic principle of hidden markov models.

8) Enlist the enzymes used in DNA replication and RNA splicing.

9) Discuss the significance of protein structure database.

10) Draw a well labelled diagram and functions of DNA.

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SECTION-B

11) Briefly discuss DNA transcription and translation process.

12) What are the different DNA sequencing techniques?

13) Explain the principle and applications of hidden markov models.

14) Briefly discuss the techniques for RNA splicing and RNA editing.

15) Explain in detail about cell structure and also describe its functions.

SECTION-C

16) Explain the biological significance of glycolysis. Discuss the krebs cycle in detail.

17) Describe the different techniques used in gene mapping. Briefly discuss about DNA restriction enzymes.

18) Discuss the principles and applications of various protein sequence databases.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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